Compact completion and uniform statistical bounds for hyperbolic geodesic sliced Wasserstein distance

Exact boundary geometry in every dimension, a correction, and a complete radial augmentation

Anonymous

4 November 2026 (version 0.1.0-candidate; unrefereed)

Contents

Abstract **2**

1. Introduction: question, answer, and scope **2**

2. Definitions and the compact-completion theorem **4**

2.1. Geometry and normalisation 4

2.2. Uniform angular moments 5

2.3. Exact constants for quadratic loss 6

3. A correction to a published backprojection claim **7**

4. Escape to infinity and the boundary geometry **7**

4.1. Loss of hyperbolic moment control 7

4.2. Sharp radial rates 8

4.3. The metric on the boundary sphere 9

5. Uniform statistical estimation **11**

5.1. Sampling error without hyperbolic moments 11

5.2. Rate optimality 12

5.3. Random-projection error 13

6. A complete augmentation using one radial comparison **13**

7. Reproducibility, limits, and research status **14**

8. Conclusion **17**

Abstract

Geodesic hyperbolic sliced Wasserstein distance is a cheap way to compare probability distributions on hyperbolic space, used for data embedded in hyperbolic geometry. We determine exactly what this distance measures. For real hyperbolic space of dimension $d \geq 2$ and every $1 \leq p < \infty$, it extends to a finite metric on all probability measures, with no moment assumption. Its metric completion is the space of probability measures on the closed Klein ball, and it metrises the weak topology there. Its exact diameter is $2C_{d,p}^{1/p}$, where $C_{d,p}$ is the pth absolute moment of the density $c_d \operatorname{sech}^{d-1} t$. Consequently it controls ordinary weak convergence but not hyperbolic transport moments, and it is exponentially insensitive to how deep points lie towards infinity. We give the three exact radial rates of approach to the ideal boundary, and the metric that the distance induces on the boundary sphere in every dimension: a square-root snowflake of the angle when $d = 2$, a logarithmically corrected angle when $d = 3$, and a metric bi-Lipschitz to the angle when $d \geq 4$. We give a counterexample to the vanishing-at-infinity assertion in Proposition 19 of Bonet, Drumetz and Courty (Journal of Machine Learning Research, 2025). We prove distribution-free $n^{-1/2}$ minimax rates for the powered loss, the matching $n^{-1/(2p)}$ rate for the distance itself, and a distribution-free high-probability bound. Adding one radial comparison gives a complete metric with the usual hyperbolic Wasserstein topology. The separation step is credited to Lyons (2014).

1. Introduction: question, answer, and scope

Context. Machine-learning methods increasingly embed data (trees, graphs, hierarchies) in hyperbolic space, where distances grow exponentially towards infinity. Comparing two such data clouds with optimal transport is expensive. *Sliced* transport makes it cheap: project both clouds onto many lines, solve the easy one-dimensional problem on each, and average. Bonet et al. (2022) (published in PMLR volume 221, 2023) introduced the geodesic version for hyperbolic space (GHSW): each "line" is a geodesic through a fixed origin o, and each point is sent to the nearest point on it. Bonet et al. (2024) (published in JMLR volume 26, 2025; all page and proposition numbers below refer to that version) extended this to Cartan–Hadamard manifolds and asked when such constructions separate distributions.

The question. Which space of distributions does GHSW really live on, and which kind of convergence does it detect? This decides what a small GHSW value guarantees in practice, how fast it can be estimated from data, and whether the space is complete.

The answer, in one paragraph. In the Klein model of hyperbolic space (the open unit ball, in which geodesics are straight chords), the projection is simply $z \mapsto \operatorname{atanh}(v \cdot z)$ (Lemma 2.1). These projections have bounded moments uniformly, even at the ideal boundary sphere. Hence GHSW extends to all probability measures on the **closed** ball, which is compact, and that compact space is exactly its completion (Theorem 2.3). Mass that escapes to infinity is therefore a legitimate limit, not a defect. Everything else follows from this picture: an exact diameter, exact rates of approach to the boundary (Theorem 4.1), an exact description of the boundary geometry in every dimension (Theorem 4.4), estimation rates that need no moment assumption at all (Section 5), and a one-term repair when transport moments matter (Section 6).

Reading guide. Section 2 sets up notation and proves the completion theorem. Section 3 corrects a published proposition. Section 4 studies what happens near infinity. Section 5 gives statistical rates. Section 6 gives the augmented metric. Section 7 states limits and reproducibility. Section 8 concludes. Table 1 summarises the results.

Table 1: Results at a glance.

Result	Statement in brief	Where
Completion	GHSW_p completes to $\mathcal{P}(\overline{B}^d)$, compact, weak topology	Thm 2.3
Diameter	exactly $2C_{d,p}^{1/p}$; $C_{d,2} = \frac{1}{2}\psi_1(\frac{d-1}{2})$	Thm 2.3, (2.14)
Correction	C_0 backprojection claim of Bonet, Drumetz and Courty (Prop. 19) fails	Prop 3.1
Radial rates	$e^{-(d-1)r}$, re^{-2pr}, e^{-2pr} across $d-1 \lessgtr 2p$	Thm 4.1
Depth insensitivity	far points at different depths are exponentially close	Cor 4.2
Boundary geometry	$\pi\alpha$ $(d=2)$; $\frac{1}{2}\alpha^2 \log\frac{1}{\alpha}$ $(d=3)$; $\asymp \alpha^2$ $(d \geq 4)$	Thm 4.4
Sampling	$\mathbb{E}D_p^p(P_n, P) \leq S_{d,p}n^{-1/2}$ for every P; high-probability version	Thm 5.1
Minimax	rate $n^{-1/2}$ for D_p^p and $n^{-1/(2p)}$ for D_p	Thm 5.2, Cor 5.3
Augmentation	adding a radial W_p term gives a complete metric with the $W_p^{\mathbb{H}}$ topology	Thm 6.1

Attribution boundary. Lyons (2014, Section 2) already uses the Klein model and the Cramér–Wold device to show that geodesic halfspace probabilities determine a hyperbolic probability measure; the separation step below applies that mechanism. The underlying link between geodesic (totally geodesic) Radon transforms in the Klein model and Euclidean Radon transforms is classical (Helgason, 2010). We claim no new injectivity theorem. Contraction under geodesic projection is in Chami et al. (2021, Proposition 3.5) and is illustrated by Bonet et al. (2022, Figure 3) and Li et al. (2025, Figure 1). Our contribution is the completion, the quantitative and statistical classification, the boundary geometry, the correction and the augmentation. A bounded search of primary sources (1–4 November 2026, including works citing Bonet et al. 2023) did not locate these results; this is not a priority certificate.

Euclidean contrast and a related release. For Euclidean sliced Wasserstein distance, completeness and Wasserstein-topology questions behave differently: Kitagawa and Takatsu (2023) [Main Theorem; published in Indiana Univ. Math. J. 75, 2026] prove completeness and topology equivalence for Euclidean sliced Monge–Kantorovich metrics, and Park and Slepčev (2025, Proposition 2.4) establish completeness in the quadratic case. A separate Evidence Press candidate (Anonymous, 2026) studies the Euclidean question of when sliced distance controls full Wasserstein distance at the critical moment; that is a different theorem, about a non-compact space. The hyperbolic geodesic construction completes differently because its transformed projections have uniformly bounded angular moments even at infinity.

Scope. All proofs are mathematical; numerical checks illustrate identities and constants and prove nothing. Results concern the exact average over all directions with a fixed origin. Finitely many directions do not separate laws (Proposition 6.2). We do not address general Cartan–Hadamard manifolds or the horospherical construction.

2. Definitions and the compact-completion theorem

2.1. Geometry and normalisation

Fix sectional curvature -1 (curvature $-\kappa^2$ divides all distances by κ; see the end of Section 2.2). Let

$$\mathbb{H}^d = \{x = (x_0, x') \in \mathbb{R}^{d+1} : x_0^2 - |x'|^2 = 1,\ x_0 > 0\}, \qquad o = (1, 0).$$

The Lorentz bilinear form is $\langle x, y\rangle_L = -x_0 y_0 + x' \cdot y'$; its restriction to each tangent space is the hyperbolic metric. We write σ for the **normalised** rotation-invariant probability measure on S^{d-1}. For $v \in S^{d-1}$ let $\gamma_v(t) = (\cosh t, \sinh t\, v)$ and let $G_v(x)$ be the signed coordinate of the nearest point to x on γ_v. The Klein coordinate map and its inverse are

$$K(x) = \frac{x'}{x_0} \in B^d, \qquad K^{-1}(z) = \frac{(1, z)}{\sqrt{1 - |z|^2}}. \tag{2.1}$$

Data stored in the Poincaré ball $y \in B^d$ convert by $K = 2y/(1 + |y|^2)$.

Write $\mathcal{P}(X)$ for Borel probability measures and $\mathcal{P}_p(\mathbb{H}^d)$ for those with finite pth moment of $d_{\mathbb{H}}(o, x)$. The geodesic sliced distance of Bonet et al. (2022, equations (18) and (21)) is

$$\mathrm{GHSW}_p(\mu, \nu)^p = \int_{S^{d-1}} W_p^p((G_v)_\# \mu, (G_v)_\# \nu)\, d\sigma(v), \tag{2.2}$$

where W_p is the real-line Wasserstein distance. We write $W_p^{\mathbb{H}}$ for the hyperbolic Wasserstein distance and SW_p for the Euclidean sliced distance of laws on $\overline{B}^d$, that is (2.2) with G_v replaced by $z \mapsto v \cdot z$.

Lemma 2.1 (projection coordinate). For every $x \in \mathbb{H}^d$,

$$G_v(x) = \operatorname{atanh}(v \cdot K(x)). \tag{2.3}$$

Proof. $\cosh d_{\mathbb{H}}(x, \gamma_v(t)) = x_0 \cosh t - (x' \cdot v) \sinh t$. Since $|x' \cdot v| < x_0$, this strictly convex function of t has its unique minimum at $\tanh t = (x' \cdot v)/x_0$. In particular $K(\exp_o(ru)) = \tanh r\, u$. □

For $z \in \overline{B}^d$ put $F_v(z) = \operatorname{atanh}(v \cdot z)$ when $|v \cdot z| < 1$, and $F_v(z) = 0$ when $|v \cdot z| = 1$. For fixed z this exceptional set has at most two directions, so it is σ-null when $d \geq 2$; by Fubini its $P \otimes \sigma$ measure is zero for every $P \in \mathcal{P}(\overline{B}^d)$. The convention therefore affects no angular integral. Define

$$D_p(P, Q)^p = \int_{S^{d-1}} W_p^p((F_v)_\# P, (F_v)_\# Q)\, d\sigma(v). \tag{2.4}$$

Measurability. For fixed P, Q, the clipped costs of (2.10) below are continuous in v and converge to the slice cost, by monotone convergence (clipped quantile differences increase to the unclipped ones), outside the exceptional directions v for which P or Q charges $\{|v \cdot z| = 1\}$. That set corresponds to boundary atoms of P or Q, so it is countable, Borel and σ-null. Changing the integrand on that null set does not change D_p, so (2.4) is a well-defined integral of a measurable function. Directions with infinite marginal pth moment are also null (Lemma 2.2).

2.2. Uniform angular moments

Put $m = d - 1$ and

$$c_d = \frac{\Gamma(d/2)}{\sqrt{\pi}\,\Gamma((d-1)/2)}, \qquad C_{d,p} = c_d \int_{\mathbb{R}} |t|^p \operatorname{sech}^{d-1} t \, dt. \tag{2.5}$$

Lemma 2.2 (angular envelope). For every finite $p \geq 1$,

$$\sup_{|z|\leq 1} \int |F_v(z)|^p \, d\sigma(v) = C_{d,p} < \infty, \tag{2.6}$$

with equality at every boundary point. If V is uniform on S^{d-1}, then $Y = \operatorname{atanh} V_1$ has density $c_d \operatorname{sech}^{d-1} t$. For $M \geq 0$,

$$\varepsilon_M^p := \sup_{|z|\leq 1} \int (|F_v(z)| - M)_+^p \, d\sigma(v) = \mathbb{E}(|Y| - M)_+^p \leq \frac{2^d c_d \Gamma(p+1)}{(d-1)^{p+1}} e^{-(d-1)M}. \tag{2.7}$$

Proof. V_1 has density $c_d(1-t^2)^{(d-3)/2}$ on $(-1,1)$; substituting $t = \tanh y$ gives the stated density of Y. By rotation invariance $\int \phi(F_v(z))\, d\sigma(v) = \mathbb{E}\phi(\operatorname{atanh}(|z|V_1))$, and $|\operatorname{atanh}(at)| \leq |\operatorname{atanh} t|$ for $0 \leq a \leq 1$ gives the suprema in (2.6) and (2.7), attained at $|z| = 1$. Finally $\operatorname{sech} y \leq 2e^{-y}$ for $y \geq 0$, so the clipped tail is at most $2c_d 2^{d-1} \int_M^\infty (t-M)^p e^{-(d-1)t}\, dt$, which is (2.7). □

Theorem 2.3 (compact completion and exact diameter). Let $d \geq 2$ and $1 \leq p < \infty$.

1. D_p is a finite metric on $\mathcal{P}(\overline{B}^d)$ and metrises its weak topology. This metric space is compact and complete.
2. Its diameter is $\operatorname{diam}(\mathcal{P}(\overline{B}^d), D_p) = 2C_{d,p}^{1/p}$, attained by antipodal boundary point masses. (2.8)
3. Under $K_\#$, both $(\mathcal{P}_p(\mathbb{H}^d), \mathrm{GHSW}_p)$ and $(\mathcal{P}(\mathbb{H}^d), \mathrm{GHSW}_p)$ embed isometrically and densely in this compact space; it is their metric completion.
4. On $\mathcal{P}(\mathbb{H}^d)$, GHSW_p metrises ordinary weak convergence. It does not control hyperbolic pth moments.

Proof. Symmetry and non-negativity are immediate; Minkowski's inequality and the one-dimensional triangle inequality give the triangle inequality. Lemma 2.2 gives

$$D_p(P, \delta_0)^p = \int\int |F_v(z)|^p \, dP(z) d\sigma(v) \leq C_{d,p}, \tag{2.9}$$

so all distances are at most $2C_{d,p}^{1/p}$; for $P = \delta_u$, $Q = \delta_{-u}$ with $|u| = 1$, oddness of F_v gives equality.

Separation. If $D_p(P, Q) = 0$, the sliced laws agree for almost every v. Applying tanh, the laws of $v \cdot z$ agree for almost every v. So the Euclidean characteristic functions of P and Q agree on every line through a direction in a full-measure, hence dense, subset of the sphere; by continuity they agree everywhere, and $P = Q$. This is the Klein/Cramér–Wold mechanism of Lyons (2014).

Continuity. Let

$$F_v^M(z) = \max\{-M, \min\{M, \operatorname{atanh}(v \cdot z)\}\}, \tag{2.10}$$

with endpoint values $\pm M$; it is jointly continuous and bounded on $S^{d-1} \times \overline{B}^d$. Let D_p^M be (2.4) with clipped slices. On $\{|v \cdot z| \le \tanh M\}$ the derivative of atanh is at most $\cosh^2 M$, so F_v^M is $\cosh^2 M$-Lipschitz in z, and a common coupling gives

$$D_p^M(P,Q) \le \cosh^2 M \, W_p^{\mathrm{E}}(P,Q), \tag{2.11}$$

with W_p^{E} the Euclidean Wasserstein distance on the compact ball. Coupling each slice with its clipping and applying Minkowski in $L^p(\sigma)$ gives, uniformly in P, Q,

$$|D_p(P,Q) - D_p^M(P,Q)| \le 2\varepsilon_M \longrightarrow 0. \tag{2.12}$$

Weak convergence on a compact set implies W_p^{E} convergence, so (2.11)–(2.12) show that weak convergence implies D_p convergence. The weak topology on $\mathcal{P}(\overline{B}^d)$ is compact, so the continuous identity map into the Hausdorff D_p topology is a homeomorphism. This proves compactness, the reverse implication and completeness.

Density. Push any P forward by $z \mapsto az$, $a \uparrow 1$. These measures converge weakly to P and live on compact subsets of the open ball, so they have finite hyperbolic moments of all orders. Formula (2.3) gives the isometry.

Part 4. If $P_n \to P$ weakly on the closed ball and P is supported in the open ball, the sequence is tight inside the open ball (use interior compacts carrying almost all the limit mass and slightly larger open neighbourhoods). On such compacts, a bounded continuous function on the open ball extends continuously to the closed ball, and the complement contributes an arbitrarily small error. The converse follows by restriction. Section 4.1 shows that moments are not controlled. □

Dimension and curvature. When $d = 1$ the two directions carry positive mass and the construction is ordinary W_p on the line; Theorem 2.3 fails. For curvature $-\kappa^2$, projection coordinates are $\kappa^{-1} F_v$, and all distances and diameters acquire a factor κ^{-1}.

2.3. Exact constants for quadratic loss

For $|s| < d - 1$, substituting $u = e^{2t}$ and using the beta integral gives

$$\mathbb{E}e^{sY} = \frac{\Gamma((d-1+s)/2)\Gamma((d-1-s)/2)}{\Gamma((d-1)/2)^2}, \tag{2.13}$$

and differentiating twice,

$$C_{d,2} = \tfrac{1}{2}\psi_1\Big(\tfrac{d-1}{2}\Big), \tag{2.14}$$

with ψ_1 the trigamma function. These are classical calculations; here they give exact geometric constants.

Table 2: Exact quadratic constants.

Dimension d	$C_{d,2}$	Diameter of D_2
2	$\pi^2/4$	π
3	$\pi^2/12$	$\pi/\sqrt{3}$
4	$\pi^2/4 - 2$	$\sqrt{\pi^2 - 8}$
5	$\pi^2/12 - 1/2$	$\sqrt{\pi^2/3 - 2}$

Dimension d	$C_{d,2}$	Diameter of D_2

3. A correction to a published backprojection claim

Bonet et al. (2024, Proposition 19, p. 24) state, for their Cartan–Hadamard Radon transform CHR on a manifold $\mathcal{M}$ with origin tangent sphere S_o: "Let $g \in C_0(\mathbb{R} \times S_o)$, then $\mathrm{CHR}^* g \in C_0(\mathcal{M})$", where C_0 denotes continuous functions vanishing at infinity. Their proof (Appendix D.3, p. 53) asserts that the set of directions $\{v : |P^v(x)| < M\}$ shrinks to the empty set as $x \to \infty$. That step holds for Euclidean linear projections, where it underlies the classical argument (Boman and Lindskog, 2009), but it fails for geodesic projections on hyperbolic space. Because $\mathbb{H}^d$ is a Cartan–Hadamard manifold, the following also refutes Proposition 19 as stated in that generality.

Proposition 3.1 (counterexample and corrected statement). For $g(t, v) = \operatorname{sech}^2 t$, the geodesic hyperbolic backprojection satisfies

$$(A^* g)(\exp_o(ru)) := \int g(G_v(\exp_o(ru)), v)\, d\sigma(v) = 1 - \frac{\tanh^2 r}{d} \longrightarrow 1 - \frac{1}{d} > 0. \tag{3.1}$$

So $g \in C_0(\mathbb{R} \times S^{d-1})$ but $A^* g \notin C_0(\mathbb{H}^d)$. More generally, for every bounded continuous g the backprojection extends continuously to $\overline{B}^d$; for $g \in C_0$ it lies in $C_0(\mathbb{H}^d)$ exactly when its boundary trace

$$u \longmapsto \int g(\operatorname{atanh}(v \cdot u), v)\, d\sigma(v) \tag{3.2}$$

vanishes identically.

Proof. $\operatorname{sech}^2(\operatorname{atanh} t) = 1 - t^2$ and $\int (v \cdot u)^2 d\sigma(v) = 1/d$. If $z_n \to z$ in $\overline{B}^d$, the integrands converge for all v except at most two endpoint directions, and boundedness permits dominated convergence. A continuous function on the compact ball vanishes at infinity on the open ball exactly when it vanishes on the boundary. □

Along $x = \exp_o(ru)$ the directional set has limiting measure $\sigma\{v : |\operatorname{atanh}(v \cdot u)| < M\} > 0$, not zero. The correction does not prevent defining the measure transform: define it directly as the pushforward of $P \otimes \sigma$ under $(z, v) \mapsto (F_v(z), v)$; its disintegrations are the sliced laws, and integration against bounded test functions gives the same adjoint identity. We have not audited which other results of Bonet et al. (2024) use the C_0 property; the sliced distances themselves are unaffected.

4. Escape to infinity and the boundary geometry

This section quantifies how the distance behaves far from the origin: it loses moment control (4.1), points approach the boundary at three possible rates (4.2), and the boundary sphere carries an explicit metric that depends on dimension (4.3).

4.1. Loss of hyperbolic moment control

Take a unit vector u, $x_n = \exp_o(n^2 u)$ and $\mu_n = (1 - n^{-1})\delta_o + n^{-1}\delta_{x_n}$. The coupling with δ_o is unique, so

$$\mathrm{GHSW}_p(\mu_n, \delta_o)^p \le \frac{C_{d,p}}{n} \longrightarrow 0, \qquad W_p^{\mathbb{H}}(\mu_n, \delta_o)^p = n^{2p-1} \longrightarrow \infty. \tag{4.1}$$

Every μ_n has all moments finite, so even on $\mathcal{P}_p(\mathbb{H}^d)$ the sliced topology is strictly weaker than the $W_p^{\mathbb{H}}$ topology, and more directions cannot help, because (4.1) uses the exact angular integral. Point masses $\delta_{\exp_o(ru)}$ are Cauchy as $r \to \infty$ and converge in the completion to the boundary point mass δ_u.

On a fixed Klein ball $|z| \le \rho < 1$ the distortion is controlled:

$$\mathrm{SW}_p(P, Q) \le D_p(P, Q) \le \frac{\mathrm{SW}_p(P, Q)}{1 - \rho^2}, \tag{4.2}$$

because $1 \le \operatorname{atanh}' \le (1 - \rho^2)^{-1}$ there and increasing maps act on quantiles by composition. A hyperbolic radius bound R gives $\rho = \tanh R$ and factor $\cosh^2 R$.

4.2. Sharp radial rates

Let $E_{d,p}(r) = D_p(\delta_{\tanh r\, u}, \delta_u)^p$ and $m = d - 1$.

Theorem 4.1 (three radial regimes). As $r \to \infty$,

$$E_{d,p}(r) \sim \begin{cases} A_{d,p} e^{-mr}, & m < 2p, \\ 2^{p+1} c_d\, r e^{-2pr}, & m = 2p, \\ 2^p c_d\, \mathrm{B}((p+1)/2, m/2 - p) e^{-2pr}, & m > 2p, \end{cases} \tag{4.3}$$

where

$$A_{d,p} = 2^{d-p-1} c_d \int_0^\infty \log^p(1 + x) x^{-(d+1)/2}\, dx. \tag{4.4}$$

Proof. By symmetry and Lemma 2.2,

$$E_{d,p}(r) = 2c_d \int_0^\infty \Delta_r(s)^p \operatorname{sech}^m s\, ds, \tag{4.5}$$

where, with $h(y) = \frac{1}{2} \log(1 + e^{2y})$,

$$\Delta_r(s) = s - \operatorname{atanh}(\tanh r \tanh s) = h(s - r) - h(-s - r). \tag{4.6}$$

If $m < 2p$, substitute $y = s - r$. The scaled integrand tends to $2^m h(y)^p e^{-my}$ and is dominated by this integrable function (extended by zero below $-r$); integrability follows from $h(y) \sim e^{2y}/2$ at $-\infty$ and $h(y) \sim y$ at $+\infty$. Dominated convergence and $x = e^{2y}$ give (4.4).

If $m > 2p$, then $e^{2r} \Delta_r(s) \to \sinh(2s)$ pointwise, and $0 \le \Delta_r(s) \le e^{2(s-r)}/2$ gives the integrable majorant $2^{-p} e^{2ps} \operatorname{sech}^m s$ after scaling. The limit is $2c_d \int_0^\infty \sinh^p(2s) \operatorname{sech}^m s\, ds = 2^p c_d \mathrm{B}((p + 1)/2, m/2 - p)$, by $t = \tanh s$ and then $q = t^2$.

If $m = 2p$, the scaled integrand without $2c_d$ is at most 2^p for $s \le r$. For $s \ge r$, put $y = s - r$ and use $\Delta_r(s) \le h(y) \le y + \frac{1}{2} \log 2$ to bound the scaled tail by an integrable function of y independent of r. So $\limsup E_{d,p}(r)/(re^{-2pr}) \le 2^{p+1} c_d$. For the lower bound, fix large M and restrict to $M \le s \le r - M$; put $a = e^{2(s-r)}$, $b = ae^{-4s}$. Then $(a - b)/(1 + a) \le \log(1 + a) - \log(1 + b) \le a - b$,

so $\Delta_r(s)/(a/2)$ lies between $(1-e^{-4M})/(1+e^{-2M})$ and 1. With $\operatorname{sech} s = 2e^{-s}/(1+e^{-2s})$, the scaled integrand is uniformly close to 2^p on an interval of length $r-2M$. Let $r\to\infty$, then $M\to\infty$. □

For quadratic loss,

$$E_{2,2}(r) \sim 4\log 2\, e^{-r}, \qquad E_{3,2}(r) \sim \frac{\pi^2}{6} e^{-2r}, \qquad E_{5,2}(r) \sim 6re^{-4r}. \tag{4.7}$$

For curvature $-\kappa^2$, replace r by κr and multiply E by κ^{-p}.

Corollary 4.2 (depth insensitivity). Let $\lambda = \min\{d-1, 2p\}$. For $r, r' \ge r_0$ on the same ray,

$$D_p(\delta_{\tanh r\, u}, \delta_{\tanh r'\, u}) \le E_{d,p}(r)^{1/p} + E_{d,p}(r')^{1/p} = O\Big((1+r_0)^{1/p} e^{-\lambda r_0/p}\Big),$$

with an implied constant depending only on d and p, while their hyperbolic distance is $|r-r'|$.

Proof. Triangle inequality through δ_u and Theorem 4.1. □

Practical meaning. Two embedded points far out along the same direction are almost indistinguishable to GHSW, however far apart they are in hyperbolic distance. A loss built only from GHSW therefore gives exponentially small signal about the depth of far-out points; Section 6 gives a cheap remedy.

4.3. The metric on the boundary sphere

The boundary sphere $\partial B^d = S^{d-1}$ sits inside the completion through $u \mapsto \delta_u$. For $p=2$ we now compute the metric this induces. By rotation invariance $D_2(\delta_u, \delta_w)^2$ depends only on the angle $\alpha \in [0,\pi]$ between u and w; call it $\beta_d(\alpha)$. In all dimensions $\beta_d(\pi) = 4C_{d,2}$.

We first record the dimension-two formula. Let $\Phi(z) = \sum_{j\ \mathrm{odd}} z^j/j^2 = \mathrm{Li}_2(z) - \frac{1}{4}\mathrm{Li}_2(z^2)$ on the closed unit disc, and use Poincaré radial parameters $q = \tanh(r/2)$, $s = \tanh(t/2)$ with angular separation α.

Proposition 4.3 (quadratic Dirac formula, $d=2$).

$$D_2(\delta_{\tanh r\, u}, \delta_{\tanh t\, w})^2 = 2\Phi(q^2) + 2\Phi(s^2) - 4\operatorname{Re}\Phi(qse^{i\alpha}), \tag{4.8}$$

and the formula extends to $q=1$ or $s=1$. In particular $\beta_2(\alpha) = \pi\alpha$ for $0 \le \alpha \le \pi$. (4.9)

Proof. Since $\tanh r = 2q/(1+q^2)$,

$$\operatorname{atanh}(\tanh r\cos\theta) = \log|1+qe^{i\theta}| - \log|1-qe^{i\theta}| = 2\sum_{j\ \mathrm{odd}} \frac{q^j\cos(j\theta)}{j}. \tag{4.10}$$

For $q<1$ the series converges absolutely; as $q\uparrow 1$ the coefficients converge in ℓ_2, so the functions converge in L^2, and the pointwise limit $\log|\cot(\theta/2)|$ off two angles identifies the L^2 limit. Parseval gives (4.8). For boundary points it gives $4\sum_{j\ \mathrm{odd}}(1-\cos j\alpha)/j^2 = \pi\alpha$, from $\sum_{j\ge1}(1-\cos j\alpha)/j^2 = \pi\alpha/2 - \alpha^2/4$ on $[0,2\pi]$ minus its even terms. □

Equation (4.8) also gives $E_{2,2}(r) = 2\{\Phi(1) + \Phi(q^2) - 2\Phi(q)\}$, an independent check on (4.7).

Theorem 4.4 (boundary geometry in every dimension).

(i) $d=2$: $\beta_2(\alpha) = \pi\alpha$, so D_2 on the boundary circle is $\sqrt{\pi\cdot\text{angle}}$.

(ii) $d = 3$: with Legendre polynomials P_l,

$$\beta_3(\alpha) = 2\sum_{l \text{ odd}} \frac{2l+1}{l^2(l+1)^2}(1 - P_l(\cos\alpha)), \qquad \beta_3(\alpha) = \tfrac{1}{2}\alpha^2 \log\tfrac{1}{\alpha} + O(\alpha^2) \quad (\alpha \to 0). \tag{4.11}$$

(iii) $d \geq 4$: for all $\alpha \in [0, \pi]$,

$$\beta_d(\alpha) \leq \kappa_d\, \alpha^2, \qquad \lim_{\alpha\to 0} \frac{\beta_d(\alpha)}{\alpha^2} = \kappa_d := \frac{d-2}{(d-1)(d-3)}, \tag{4.12}$$

and $\beta_d(\alpha) \geq c\,\alpha^2$ for some $c = c(d) > 0$. Thus for $d \geq 4$ the boundary sphere with D_2 is bi-Lipschitz equivalent to the round sphere with its angle metric; for $d = 3$ it is not (the ratio D_2/α grows like $\sqrt{\frac{1}{2}\log(1/\alpha)}$); for $d = 2$ it is the square-root snowflake of the angle.

Proof. Write $f(t) = \operatorname{atanh} t$, so $\beta_d(\alpha) = \|f_u - f_w\|^2_{L^2(\sigma)}$ with $f_u(v) = f(v \cdot u)$; $f_u \in L^2(\sigma)$ by Lemma 2.2.

(ii) In $L^2([-1,1])$, $f = \sum_l a_l P_l$ with $a_l = \frac{2l+1}{2}\int_{-1}^1 f P_l$. Since f is odd, $a_l = 0$ for even l; for odd l, $\int_{-1}^1 \operatorname{atanh}(t) P_l(t)\, dt = 2/(l(l+1))$ (integrate by parts using $(1-t^2)P_l' = l(P_{l-1} - tP_l)$, or by Legendre's formula; the values are also checked numerically), so $a_l = (2l+1)/(l(l+1))$. On S^2 the addition theorem gives $\int P_l(v\cdot u)P_k(v\cdot w)\, d\sigma(v) = \delta_{lk} P_l(u\cdot w)/(2l+1)$. Hence $\int f_u f_w\, d\sigma = \sum_{l \text{ odd}} a_l^2 P_l(\cos\alpha)/(2l+1)$, and $\beta_3 = 2(\|f_u\|^2 - \int f_u f_w)$ is the series in (4.11). For the asymptotic, let $N = \lfloor 1/\alpha \rfloor$. Taylor expansion at $x = 1$ with $P_l'(1) = l(l+1)/2$ and $\max_{[-1,1]} |P_l''| = P_l''(1) = O(l^4)$ gives $1 - P_l(\cos\alpha) = l(l+1)\alpha^2/4 + O(l^4\alpha^4)$ uniformly in l. Summing over odd $l \leq N$ gives $(\alpha^2/2)\sum_{l \text{ odd}\leq N}(2l+1)/(l(l+1)) + O(\alpha^4 N^2) = \frac{1}{2}\alpha^2 \log N + O(\alpha^2)$, using $\sum_{l \text{ odd}\leq N} 2/l = \log N + O(1)$. The terms $l > N$ contribute at most $4\sum_{l>N}(2l+1)/(l^2(l+1)^2) = O(N^{-2}) = O(\alpha^2)$. Since $\log N = \log(1/\alpha) + O(\alpha)$, (4.11) follows.

(iii) Let R_θ rotate a two-plane containing u and w (any such plane if $w = \pm u$) by angle θ, with $R_\alpha u = w$, and let $e_\theta = \partial_\theta R_\theta u$, a unit vector orthogonal to $R_\theta u$. For σ-almost every v the arc $\{R_\theta u : 0 \leq \theta \leq \alpha\}$ avoids $\pm v$ (the union over θ of the two bad directions is contained in a great circle, which is null when $d \geq 3$), so $\theta \mapsto f(v \cdot R_\theta u)$ is C^1 on $[0, \alpha]$ with derivative $X_\theta(v) = (v \cdot e_\theta)/(1 - (v \cdot R_\theta u)^2)$. Hence $f_w - f_u = \int_0^\alpha X_\theta\, d\theta$ pointwise almost everywhere; as $\int_0^\alpha \|X_\theta\|\, d\theta < \infty$ (below), the same identity holds as a Bochner integral in $L^2(\sigma)$. By Cauchy–Schwarz, Fubini and rotation invariance,

$$\beta_d(\alpha) = \Big\| \int_0^\alpha X_\theta\, d\theta \Big\|^2 \leq \alpha \int_0^\alpha \|X_\theta\|^2 d\theta = \alpha^2 \|X_0\|^2.$$

Conditioning on $v \cdot u = t$, the component $v \cdot e_0$ has conditional second moment $(1-t^2)/(d-1)$, so

$$\|X_0\|^2 = \frac{1}{d-1}\,\mathbb{E}\frac{1}{1-V_1^2} = \frac{c_d}{d-1}\int_{-1}^1 (1-t^2)^{(d-5)/2} dt = \frac{c_d\, \mathrm{B}(\frac{1}{2}, \frac{d-3}{2})}{d-1} = \frac{d-2}{(d-1)(d-3)} = \kappa_d,$$

finite exactly when $d \geq 4$. This proves the upper bound. For the limit, $X_\theta = U_\theta X_0$ where $U_\theta g = g \circ R_\theta^{-1}$ is a strongly continuous unitary group on $L^2(\sigma)$; hence $\alpha^{-1}\int_0^\alpha X_\theta\, d\theta \to X_0$ in L^2 and $\beta_d(\alpha)/\alpha^2 \to \|X_0\|^2$. Finally, β_d is continuous on $[0, \pi]$ (Theorem 2.3) and positive for $\alpha > 0$ (separation), and $\beta_d(\alpha)/\alpha^2 \to \kappa_d > 0$, so $\inf_{0<\alpha\leq\pi} \beta_d(\alpha)/\alpha^2 > 0$. Part (i) is (4.9). □

This mirrors the radial trichotomy of Theorem 4.1 for $p = 2$: the critical dimension there is $d = 5$, here it is $d = 3$, and both reflect integrability of a power of $1/(1 - V_1^2)$ under the angular law.

5. Uniform statistical estimation

The bounded geometry of Section 2 has a practical payoff: GHSW can be estimated from samples at rates that hold uniformly over **all** distributions, with no moment or support condition.

5.1. Sampling error without hyperbolic moments

Let $P_n = n^{-1}\sum_{i=1}^n \delta_{Z_i}$ for independent $Z_i \sim P$. Define the one-sided envelope $q_d(t) = \mathbb{P}(Y > t) = c_d \int_t^\infty \operatorname{sech}^{d-1} s\, ds$ for $t \geq 0$ and

$$S_{d,p} = p2^p \int_0^\infty t^{p-1}\sqrt{q_d(t)}\, dt < \infty. \tag{5.1}$$

Theorem 5.1 (uniform empirical bounds). For every $P \in \mathcal{P}(\overline{B}^d)$:

(i) in expectation,

$$\mathbb{E}D_p(P_n, P)^p \leq \frac{S_{d,p}}{\sqrt{n}}, \qquad S_{d,p} \leq \frac{2^{2p}\Gamma(p+1)}{(d-1)^p}\sqrt{\frac{2^{d-1}c_d}{d-1}}; \tag{5.2}$$

(ii) with high probability: for every $0 < \delta < 1$,

$$\mathbb{P}\left[D_p(P_n, P)^p > \frac{S_{d,p} + 2^{p-1/2}C_{d,p}\sqrt{\log(1/\delta)}}{\sqrt{n}}\right] \leq \delta. \tag{5.3}$$

In particular $\mathbb{E}D_p(P_n, P) \leq S_{d,p}^{1/p} n^{-1/(2p)}$. The constants depend only on d, p.

Proof. For real laws α, β with distribution functions F_α, F_β, let $h(t) = \operatorname{sgn}(t)|t|^p$. Since $|x-y|^p \leq 2^{p-1}|h(x) - h(y)|$, the monotone coupling gives

$$W_p^p(\alpha, \beta) \leq p2^{p-1}\int_{\mathbb{R}} |t|^{p-1}|F_\alpha(t) - F_\beta(t)|\, dt, \tag{5.4}$$

because $h'(t) = p|t|^{p-1}$ and the probability that a monotone-coupled pair straddles t is $|F_\alpha(t) - F_\beta(t)|$. Write H_v and $H_{n,v}$ for the distribution functions of the slices $(F_v)_\# P$ and $(F_v)_\# P_n$. Apply (5.4) to each slice and integrate over v:

$$D_p(P_n, P)^p \leq U_n := p2^{p-1}\int_{S^{d-1}}\int_{\mathbb{R}} |t|^{p-1}|H_{n,v}(t) - H_v(t)|\, dt\, d\sigma(v),$$

a measurable function of $(Z_1, \ldots, Z_n)$ by Tonelli, since the integrand is jointly measurable.

(i) $\mathbb{E}|H_{n,v}(t) - H_v(t)| \leq \sqrt{H_v(t)(1 - H_v(t))/n}$. For $t > 0$, Jensen's inequality over directions bounds the angular average of $\sqrt{1 - H_v(t)}$ by the square root of the averaged right tail, which is at most $q_d(t)$ by rotation invariance and $|z| \leq 1$; the same holds for left tails at $-t$. Tonelli gives $\mathbb{E}U_n \leq S_{d,p}/\sqrt{n}$. The envelope $q_d(t) \leq c_d 2^{d-1}e^{-(d-1)t}/(d-1)$ gives the bound on $S_{d,p}$.

(ii) Replacing one observation z by z' changes each $H_{n,v}$ by $1/n$ on the interval between $F_v(z)$ and $F_v(z')$ only, so it changes U_n by at most $\frac{2^{p-1}}{n}\int |h(F_v(z)) - h(F_v(z'))|\, d\sigma(v) \leq \frac{2^{p-1}}{n} \cdot 2C_{d,p}$ by Lemma 2.2. McDiarmid's bounded-differences inequality (McDiarmid, 1989) gives $\mathbb{P}[U_n - \mathbb{E}U_n > \epsilon] \leq \exp(-2\epsilon^2 n/(2^p C_{d,p})^2)$; take $\epsilon = 2^{p-1/2}C_{d,p}\sqrt{\log(1/\delta)/n}$ and use (i). □

The argument follows the slicing method of Nadjahi et al. (2020); the new ingredient is the uniform angular tail envelope. The bounds of Bonet et al. (2022, Proposition B.6) and Bonet et al. (2024, Proposition 29) contain hyperbolic moment factors; for the geodesic hyperbolic construction these factors are unnecessary. For two independent samples, the triangle inequality gives

$$\mathbb{E}|D_p(P_n, Q_k) - D_p(P, Q)| \le S_{d,p}^{1/p}(n^{-1/(2p)} + k^{-1/(2p)}). \tag{5.5}$$

5.2. Rate optimality

For distinct $a, b \in \overline{B}^d$ put $L = D_p(\delta_a, \delta_b) > 0$. For their equal mixture, one-dimensional transport gives exactly

$$\mathbb{E}D_p(P_n, P)^p = L^p\,\mathbb{E}\left|\frac{B_n}{n} - \frac{1}{2}\right|, \qquad B_n \sim \mathrm{Bin}(n, 1/2), \tag{5.6}$$

so $\sqrt{n}\,\mathbb{E}D_p(P_n, P)^p \to L^p/\sqrt{2\pi}$ (5.7): the exponent in Theorem 5.1 cannot be improved even for bounded interior support.

Theorem 5.2 (minimax rate for the powered loss). For fixed $d \ge 2$ and $1 \le p < \infty$,

$$\frac{C_{d,p}}{8\sqrt{n}} \le \inf_{\widehat{P}} \sup_{P \in \mathcal{P}(\overline{B}^d)} \mathbb{E}_P D_p(\widehat{P}, P)^p \le \frac{S_{d,p}}{\sqrt{n}}, \tag{5.8}$$

the infimum being over measurable estimators with values in $\mathcal{P}(\overline{B}^d)$ (weak Borel σ-algebra) based on n observations. The same constants hold for the parameter class $\mathcal{P}_p(\mathbb{H}^d)$ under the Klein identification, whether estimators take values in the completion or in that class.

Proof. The upper bound is Theorem 5.1. For the lower bound [the two-point method; see Tsybakov (2009), Chapter 2], choose distinct a, b, put $\eta = 1/(4\sqrt{n})$ and

$$P_\pm = (1/2 \pm \eta)\delta_a + (1/2 \mp \eta)\delta_b.$$

Then $\Delta^p := D_p(P_+, P_-)^p = 2\eta L^p$. The one-observation Hellinger affinity is $\sqrt{1 - 4\eta^2}$, so with $\mathrm{TV}(P, Q) = \sup_A |P(A) - Q(A)|$ and Cauchy–Schwarz,

$$\mathrm{TV}(P_+^{\otimes n}, P_-^{\otimes n}) \le \sqrt{1 - (1 - 4\eta^2)^n} \le 2\sqrt{n}\eta = \tfrac{1}{2}.$$

Classify the data by whichever of $P_\pm$ is closer to $\widehat{P}$. Because D_p is a genuine metric, a classification error forces $D_p(\widehat{P}, P_\pm) \ge \Delta/2$, and the two error probabilities sum to at least $1 - \mathrm{TV} \ge \frac{1}{2}$. Hence

$$\max_{\pm} \mathbb{E}_{P_\pm} D_p(\widehat{P}, P_\pm)^p \ge \frac{\Delta^p}{2^{p+2}} = \frac{L^p}{2^{p+3}\sqrt{n}}. \tag{5.9}$$

Antipodal boundary points give $L^p = 2^p C_{d,p}$ and (5.8). For $\mathcal{P}_p(\mathbb{H}^d)$ use $\pm\rho u$ and let $\rho \uparrow 1$. □

Corollary 5.3 (minimax rate for the distance itself). For fixed $d \ge 2$ and $1 \le p < \infty$,

$$\frac{C_{d,p}^{1/p}}{4\,(2\sqrt{n})^{1/p}} \le \inf_{\widehat{P}} \sup_{P} \mathbb{E}_P D_p(\widehat{P}, P) \le S_{d,p}^{1/p}\, n^{-1/(2p)}, \tag{5.10}$$

so $n^{-1/(2p)}$ is the minimax rate for unpowered D_p, over $\mathcal{P}(\overline{B}^d)$ and over $\mathcal{P}_p(\mathbb{H}^d)$.

Proof. The upper bound is Jensen's inequality and Theorem 5.1. In the proof of Theorem 5.2, a classification error forces loss $D_p \geq \Delta/2$, so $\max_\pm \mathbb{E}_\pm D_p(\widehat{P}, P_\pm) \geq \Delta/8$ with $\Delta = (L^p/(2\sqrt{n}))^{1/p}$; for antipodal boundary points $\Delta = 2C_{d,p}^{1/p}(2\sqrt{n})^{-1/p}$. □

Constants. The rates are optimal; the constants are not. The ratio of upper to lower constants in (5.8) is about 21 ($d = 2$) to 19 ($d = 10$) for $p = 1$, and about 82 to 62 for $p = 2$ (computed numerically). Neither result gives moment-uniform estimation of $W_p^{\mathbb{H}}$, which is impossible by Section 4.1.

5.3. Random-projection error

Given independent uniform directions $V_1, \ldots, V_\ell$, let

$$\widehat{D}_{p,\ell}^{\,p}(P, Q) = \frac{1}{\ell} \sum_{j=1}^{\ell} W_p^p((F_{V_j})_\# P, (F_{V_j})_\# Q).$$

Proposition 5.4. For every **fixed** pair $P, Q \in \mathcal{P}(\overline{B}^d)$, the estimator is unbiased and

$$\mathbb{E}[\widehat{D}_{p,\ell}^{\,p} - D_p^p]^2 \leq \frac{2^{2p} C_{d,2p}}{\ell}. \tag{5.11}$$

Proof. Unbiasedness is immediate, so the mean-square error is the variance. A product coupling bounds the one-direction cost by $2^{p-1}(M_p^P(v) + M_p^Q(v))$ with $M_p^P(v) = \int |F_v(z)|^p dP(z)$. Squaring with $(a+b)^2 \leq 2(a^2+b^2)$ and $(M_p^P)^2 \leq M_{2p}^P$ and integrating gives second moment at most $2^{2p} C_{d,2p}$; independence divides by ℓ. □

The bound is uniform over fixed pairs, not simultaneous over pairs chosen after seeing the directions: for any realised directions, $P_\rho = \delta_{\rho V_1}$ and $Q = \delta_0$ give $\widehat{D}_{p,\ell}^{\,p} \geq \ell^{-1} \operatorname{atanh}(\rho)^p \to \infty$ while $D_p^p \leq C_{d,p}$.

6. A complete augmentation using one radial comparison

The bounded metric is the right tool when weak convergence is the target. When transport moments or depth matter (Section 4.1 and Corollary 4.2), one radial comparison repairs it. Set

$$R(x) = d_{\mathbb{H}}(o, x), \qquad \mathcal{D}_p(\mu, \nu) = \left\{ \mathrm{GHSW}_p(\mu, \nu)^p + W_p^p(R_\#\mu, R_\#\nu) \right\}^{1/p}. \tag{6.1}$$

Theorem 6.1 (complete radial augmentation). On $\mathcal{P}_p(\mathbb{H}^d)$, $\mathcal{D}_p$ is a complete metric,

$$\mathcal{D}_p(\mu, \nu) \leq 2^{1/p} W_p^{\mathbb{H}}(\mu, \nu), \tag{6.2}$$

and for any sequence and limit in this space

$$\mathcal{D}_p(\mu_n, \mu) \to 0 \iff W_p^{\mathbb{H}}(\mu_n, \mu) \to 0. \tag{6.3}$$

Proof. The first component is a metric and the second a pseudometric, so their ℓ_p combination is a metric. Both R and G_v are 1-Lipschitz: for G_v, write $a = x' \cdot v$, $b = x_0$, $f = b^2 - a^2 \geq 1$;

the vector $H = (a, bv)$ is tangent at x with Lorentz squared norm f and represents $b\,da - a\,db$, so $\nabla G_v = H/f$ has norm $f^{-1/2} \le 1$ (equivalently, nearest-point projection onto a geodesic is 1-Lipschitz in a CAT(-1) space). Pushing forward a common coupling gives (6.2).

If $\mathcal{D}_p(\mu_n, \mu) \to 0$, Theorem 2.3 gives weak convergence, and radial W_p convergence gives uniform integrability of R^p. With a Skorohod coupling $X_n \to X$ a.s. and $d_{\mathbb{H}}(X_n, X)^p \le 2^{p-1}\{R(X_n)^p + R(X)^p\}$, uniform integrability gives $W_p^{\mathbb{H}}$ convergence [this is the standard characterisation of Wasserstein convergence]. The converse is (6.2).

For completeness, let (μ_n) be $\mathcal{D}_p$-Cauchy. The Klein pushforwards converge in D_p to some $P \in \mathcal{P}(\overline{B}^d)$ and the radial laws converge in W_p to some $\eta \in \mathcal{P}_p([0, \infty))$; in particular the radial laws are uniformly tight. Then P charges no boundary point: continuous cutoffs supported in $\{|z| > \tanh R_0\}$ bound the limiting boundary mass by a uniformly small radial tail. Write $P = K_{\#}\mu$; Theorem 2.3 gives $\mu_n \Rightarrow \mu$, so $R_{\#}\mu_n \Rightarrow R_{\#}\mu$, hence $\eta = R_{\#}\mu$, $\mu \in \mathcal{P}_p$, and both terms of (6.1) tend to zero. □

For equal-size empirical measures this adds one sort of the radii and one one-dimensional transport, preserving the cost $O(\ell nd + \ell n \log n)$ for ℓ directions (an operation count, not a benchmark). Radii should be stored directly when Klein coordinates would round to unit norm.

Comparing only scalar moments is not enough: $\mu_n = (1 - n^{-p})\delta_o + n^{-p}\delta_{\exp_o(nu)}$ has radial pth moment one but converges weakly to δ_o, so GHSW plus a moment-norm difference would be Cauchy without a limit in $\mathcal{P}_p$. The augmentation restores convergence and completeness, not equality of distances: opposite point masses $\delta_{\exp_o(\pm ru)}$ have equal radial laws and augmented distance at most $2C_{d,p}^{1/p}$, while their hyperbolic Wasserstein distance is $2r$. The distribution-free rates of Section 5 belong to the bounded part; the radial part needs tail assumptions.

Proposition 6.2 (finitely many directions do not separate). For every finite set $v_1, \dots, v_J$ ($d \ge 2$) there are distinct laws with smooth compactly supported Klein densities with identical laws of F_{v_j} for all j and identical radial laws.

Proof. Choose nonzero $w_j \perp v_j$ and a nonzero radial $\varphi \in C_c^\infty(B^d)$ supported in a smaller ball. Let $f = \prod_j (w_j \cdot \nabla)\varphi$, adding one more directional derivative if needed to make the order odd. Each directional derivative is injective on C_c^∞, so $f \neq 0$. Being odd, f integrates to zero against every bounded radial function; and since $f = (w_j \cdot \nabla)\psi_j$ with $\psi_j \in C_c^\infty$, integrating along w_j-fibres shows that the $v_j \cdot z$ pushforward of f vanishes, and so does its atanh image. With a smooth radial probability density q positive on the support of f, the densities $q \pm \epsilon f$ work for small $\epsilon > 0$. □

7. Reproducibility, limits, and research status

The scripts check the projection formula against direct minimisation, the angular constants by quadrature, the dimension-two Fourier/dilogarithm identity, the backprojection counterexample, the radial regimes and the exact binomial errors (`code/verify_and_plot.py`), and the new boundary-sphere results and a near-critical radial constant (`code/verify_extensions.py`). They stop with an error if any check fails, also under `python -O`. Floating-point agreement is a sanity check, not a proof, and no interval certificates are claimed.

The recorded run checked 36 projection instances, six quadratic angular constants, 12 backprojection evaluations, seven dilogarithm/boundary comparisons, 24 radial cases over eight (d, p) pairs and ten binomial sums. The largest coordinate discrepancy against scalar minimisation was below 10^{-7}; dilogarithm discrepancies were below 10^{-12}; trigamma constants agreed to 5×10^{-16}. The extension script confirms the Legendre coefficients, matches the $d = 3$ series to

direct quadrature, shows $\beta_d(\alpha)/\alpha^2$ approaching κ_d for $d = 4, 5, 6$, and shows the slow approach of $\beta_3(\alpha)/(\alpha^2 \log(1/\alpha))$ towards $1/2$.

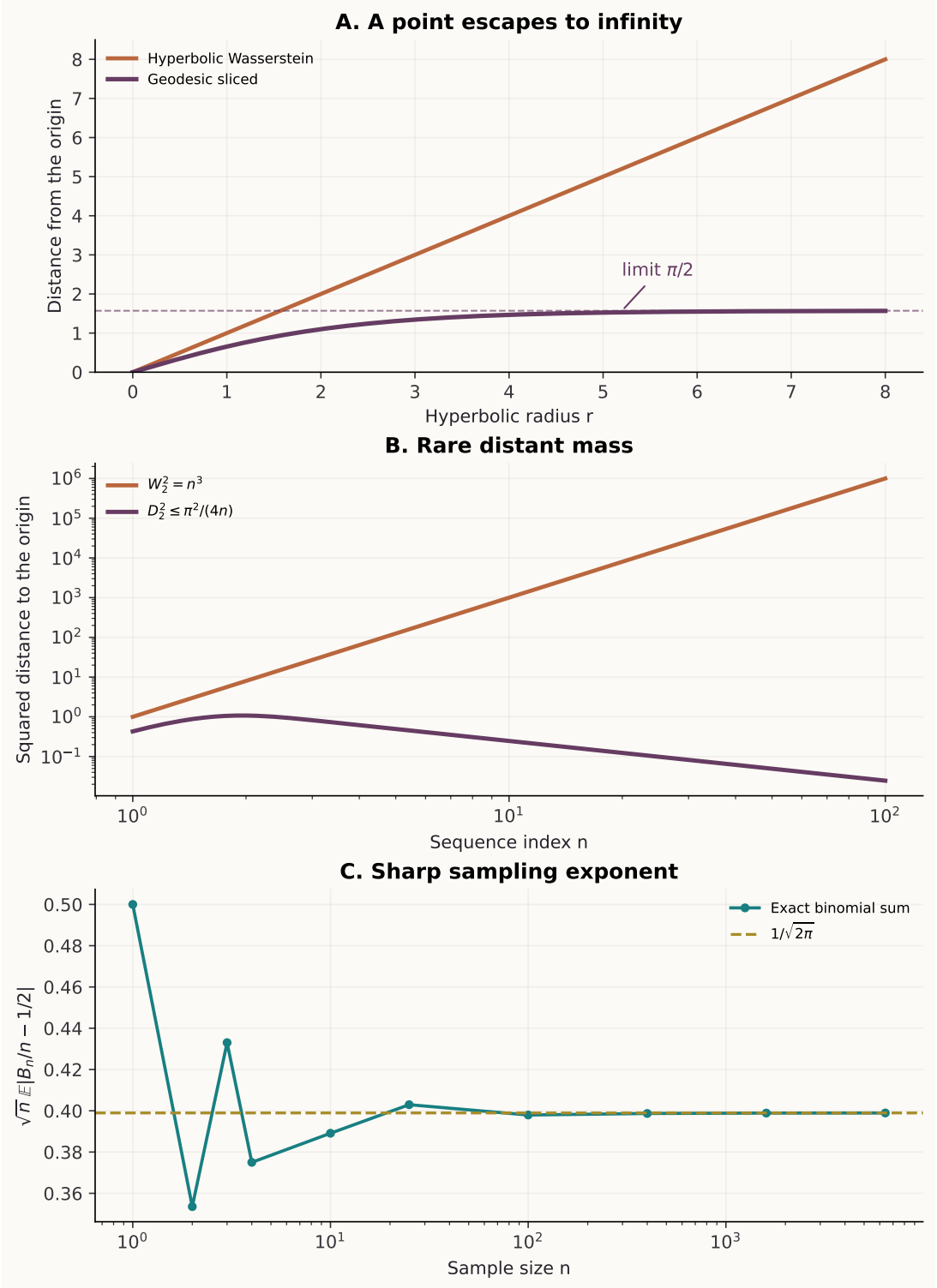


Figure 1: Geometry and sampling checks. Panel A compares a point mass at radius r with the origin ($d = 2$, $p = 2$). Panel B uses the sequence (4.1) with $p = 2$. Panel C evaluates the exact binomial imbalance (5.6); parity explains the small alternation.

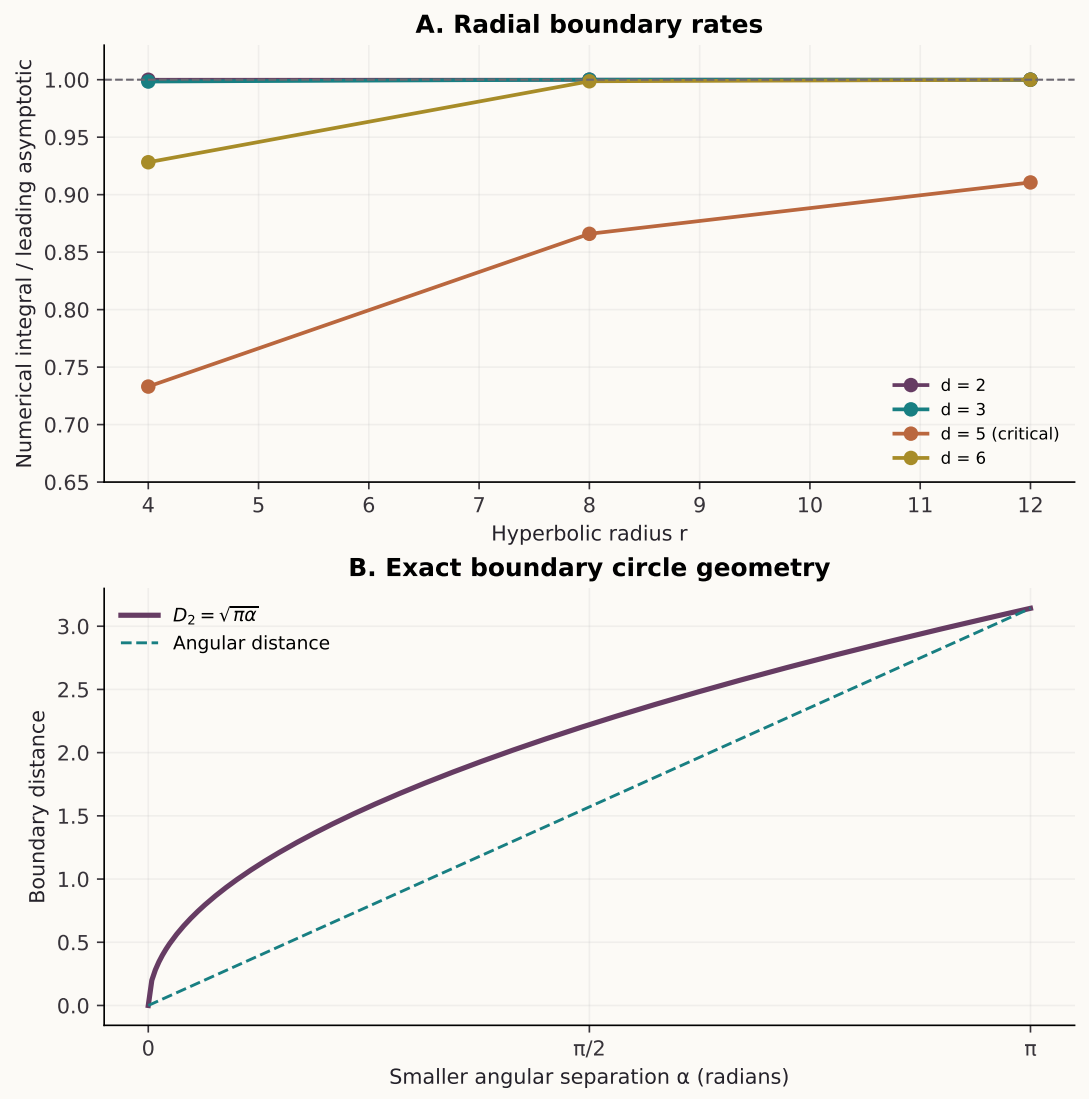


Figure 2: Boundary checks for quadratic loss. Panel A divides (4.5) by its leading asymptotic; the critical dimension $d = 5$ converges more slowly. Panel B shows the boundary metric (4.9) for $d = 2$; the dashed angular distance is a reference only.

Limits. The theorems concern real hyperbolic space, a fixed origin, the uniform angular measure, $d \geq 2$ and finite p. Changing the origin changes the metric (by an isometry of the whole picture), so results hold for each fixed origin but data-dependent origins are not covered. Other angular measures, the horospherical construction, general Cartan–Hadamard manifolds, sharp minimax constants and sampling rates for the augmented metric remain open.

Status. This is an unrefereed candidate. It has had AI-assisted review (five simulated reviewers, an independent adversarial review and a clean replay of the code), not external human peer review or formal verification. The literature search is bounded and cannot certify priority. This version (0.1.0-candidate) is archived at Zenodo with DOI 10.5281/zenodo.23938270 (all versions: 10.5281/zenodo.23938269); code and data are at https://github.com/ipitchford/hyperbolic-sliced-wasserstein-completion, and a plain-English summary is at https://evidencepress.org/releases/hyperbolic-sliced-wasserstein-completion/.

8. Conclusion

Geodesic hyperbolic sliced Wasserstein distance is, exactly, a metric for weak convergence on the compact closed Klein ball. This one fact explains its behaviour: it is bounded with an exact diameter, it sees mass escaping to infinity as converging to the boundary, it barely distinguishes depths far from the origin, its boundary sphere has a dimension-dependent geometry that we computed in every dimension, and it can be estimated at optimal distribution-free rates. Users who need transport-moment control should add the one radial term of Theorem 6.1. The correction to Proposition 19 of Bonet et al. (2024) shows that the vanishing-at-infinity step does not transfer from Euclidean to hyperbolic projections.

References

Anonymous. Sliced Wasserstein distance at the critical moment: the noninteger case, 2026. Version 1.1.0-candidate, Evidence Press research release.

Jan Boman and Filip Lindskog. Support Theorems for the Radon Transform and Cramér-Wold Theorems. *Journal of Theoretical Probability*, 22(3):683–710, 2009. doi: 10.1007/s10959-008-0151-0.

Clément Bonet, Laetitia Chapel, Lucas Drumetz, and Nicolas Courty. Hyperbolic Sliced-Wasserstein via Geodesic and Horospherical Projections. arXiv:2211.10066, 2022. arXiv comment: Accepted at the TAG-ML 2023 ICML Workshop.

Clément Bonet, Lucas Drumetz, and Nicolas Courty. Sliced-Wasserstein Distances and Flows on Cartan-Hadamard Manifolds. arXiv:2403.06560, 2024.

Ines Chami, Albert Gu, Dat Nguyen, and Christopher Ré. HoroPCA: Hyperbolic Dimensionality Reduction via Horospherical Projections. arXiv:2106.03306, 2021. arXiv comment: ICML 2021.

Sigurdur Helgason. *Integral Geometry and Radon Transforms*. Springer New York, 2010. doi: 10.1007/978-1-4419-6055-9.

Jun Kitagawa and Asuka Takatsu. Sliced optimal transport: is it a suitable replacement? arXiv:2311.15874, 2023. arXiv comment: 26 pages, added results on non bi-Lipschitz equivalence when q is infinity, split off results on disintegrated metrics (to be posted separately), changed title. Comments welcome!

Tao Li, Cheng Meng, Hongteng Xu, and Jun Zhu. Efficient Variants of Wasserstein Distance in Hyperbolic Space via Space-Filling Curve Projection. *IEEE Transactions on Neural Networks and Learning Systems*, 36(9):16814–16824, 2025. doi: 10.1109/tnnls.2025.3551275.

Russell Lyons. Hyperbolic space has strong negative type. *Illinois Journal of Mathematics*, 58 (4):1009–1013, 2014. doi: 10.1215/ijm/1446819297.

Colin McDiarmid. On the method of bounded differences. In *Surveys in Combinatorics, 1989*, pages 148–188. Cambridge University Press, 1989. doi: 10.1017/cbo9781107359949.008.

Kimia Nadjahi, Alain Durmus, Lénaïc Chizat, Soheil Kolouri, Shahin Shahrampour, and Umut Şimşekli. Statistical and Topological Properties of Sliced Probability Divergences. arXiv:2003.05783, 2020. arXiv comment: Published at NeurIPS 2020 (Spotlight).

Sangmin Park and Dejan Slepčev. Geometry and analytic properties of the sliced Wasserstein space. *Journal of Functional Analysis*, 289(7):110975, 2025. doi: 10.1016/j.jfa.2025.110975.

Alexandre B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer Series in Statistics. Springer New York, 2009. doi: 10.1007/b13794.